

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of M. Pharm
(Pharmaceutical Technology and Pharmaceutics (Drug Regulatory Affairs))
University Theory Examination March-April 2018
PHCC001 BASIC ANALYTICAL TECHNIQUES

Date: 26.03.2018, Monday Time: 10.00 a.m. to 01.00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** Write a note on Gas chromatography (GC). **05**
 B) Enlist various steps involved in HPTLC method development and explain any two steps in brief. **05**
- Q.2** **A)** Discuss in brief the instrumentation of Mass spectrometer. **05**
 B) Write a note on fragmentation rules. **05**
- OR**
- Q.2** **A)** Write a note on detectors used in HPLC. **05**
 B) Give classification of chromatography based on mechanism of separation. **05**
- Q.3** **Answer the following (Any three)** **15**
 (A) Give applications of HPLC.
 (B) Write a note on DSC.
 (C) Give advantages and disadvantages of HPTLC.
 (D) Write a note on detectors used in GC.

SECTION-II

- Q.4** **(A)** Discuss the instrumentation of UV-Visible spectrophotometer. **05**
 (B) Enlist various mass analyser. Explain any one in detail. **05**
- Q.5** **(A)** Discuss application of NMR spectroscopy. **05**
 (B) Explain the principle of Electrophoretic techniques. **05**
- OR**
- Q.5** Write note on different microscopic techniques and their applications in pharmacy. **10**
- Q.6** **Answer the following (Any three)** **15**
 (A) Write note on Reference standard compound.
 (B) Write note on Michaelson Interferometer in FT-IR.
 (C) Brief note on X-Ray diffraction method.
 (D) Write note on methods for identification of impurities.
